Collaborative Self-Supervised Evolution for Few-Shot Remote Sensing Scene Classification

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***Abstract*— Self-supervised learning, which leverages unlabeled data to learn useful feature representations by constructing auxiliary tasks, has been widely explored in few-shot scene classification to improve the feature representation and generalization capabilities of deep models in scarce data. However, most of the current related work adopts specific self-supervised auxiliary tasks (SSATs) for combinatorial improvement, and does not explore the intrinsic connection between different pretext tasks. In practice, the linkage of SSATs is complex, and the optimization of task-sharing parameters by minimizing linear combinations of losses can be conflicting. In addition, although a single combination of SSAT can improve certain performance on the baseline, it is not the personalized optimal solution on various remote sensing datasets with diverse properties. In this article, we propose a collaborative self-supervised evolution (so-called CSENet) framework for few-shot remote sensing scene classification to automatically search for appropriate weights in balancing the task conflicts. In contrast to most existing methods, which consider all SSATs to be equally efficacious or fixed-weighted for the few-shot main task, CSENet achieves autonomous co-evolutionary optimization by encoding arbitrary self-supervised weights. Specifically, the complex self-supervised combinations for different remote sensing data are transformed into an evolutionary optimization problem, where chromosomes with weighting variables obtain the optimal combination with genetic operators. Based on the transfer learning few-shot training paradigm, CSENet first efficiently searches for optimal self-supervised combinations with potential by the proposed automatic collaborative evolution strategy and automatically adjusts the weights without manual settings. Importantly, CSENet provides both inductive and transductive inference, and supports the embedding of arbitrary SSATs. The effectiveness of the proposed framework is demonstrated by state-of-the-art (SOTA) results on three benchmark datasets.**

***Index Terms*— Collaborative evolution strategy, few-shot learning, remote sensing scene classification, self-supervised learning.**

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I. Introduction

SCENE classification is to utilize the feature and spatial information in the remote sensing scenario to divide and label the ground cover into predefined categories [1]. It has become an important direction to analyze the ground cover information and is widely used in environmental monitoring [2], urban planning [3], and agricultural production [4]. However, the current deep learning-based scene classification methods usually require a large number of labeled samples for training, and the models cannot quickly generalize to unseen categories [5], [6].

In recent years, inspired by the ability of humans to recognize never-before-seen categories from only a small amount of experience, few-shot learning has been widely explored in remote sensing scene classification and achieved remarkable effects [7]. In order to improve the generalization of the model to new data, few-shot scene classification divides the dataset into the base dataset and the novel dataset, where the categories are independent of each other. In accordance with the training approach, few-shot scene classification methods can be categorized into meta-learning-based methods [8], [9], [10], [11], [12], [13], [14], [15], [16], [17] and transfer learning-based methods [18], [19], [20], [21], [22], [23]. Meta-learning-based methods utilize episodic-based training to equip the model with the ability to rapidly capture new concepts. This type of methods can be further grouped into optimization-based methods [9], [14], [15], [16] and metric-based methods [8], [10], [11], [12], [13]. Optimization-based methods aim to obtain a generalized initial model with an iterative approach, while metric-based methods classify samples by metricizing their distance or similarity under the target feature space. Recently, the effectiveness of transfer learning in few-shot scenarios has been extensively explored in remote sensing scenario classification. Transfer learning-based approaches generally train the feature extractor on the base dataset and transfer it to the novel dataset for testing [24].

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TABLE I
SUMMARY OF DIFFERENT SELF-SUPERVISED FEW-SHOT LEARNING METHODS, WHERE "*" DENOTES THAT IT IS SPECIFICALLY DESIGNED FOR THE FEW-SHOT REMOTE SENSING SCENE CLASSIFICATION TASK

Year	Method	Self-Supervised Tasks	Weights	Weight Setting	Training Paradigm	Inference
2020	S2M2 [19]	Rotation, Exemplar	1	Manual	Transfer Learning-based	Inductive
2020	Su *et al.* [26]	Rotation, Jigsaw Puzzle	1	Manual	Metric-based	Inductive
2021	Meta-FSEO* [27]	Contrastive Learning	0.3	Manual	Optimization-based	Inductive
2021	Chen *et al.* [28]	Contrastive Learning	1	Manual	Metric-based	Transductive
2021	CSS [29]	Contrastive Learning	1	Manual	Metric-based	Inductive
2021	TIM [30]	Contrastive Learning	1	Manual	Metric-based	Transductive
2022	TSC* [31]	Task-Specific Contrastive Learning	0.6	Manual	Metric-based	Inductive
2022	Ji *et al.*$_{(TGRS)}$* [20]	Rotation, Contrastive Learning	1	Manual	Transfer Learning-based	Inductive
2022	SCL-MLNet* [32]	Contrastive Learning	0.1	Manual	Metric-based	Inductive
2022	SSMR* [33]	Contrastive Learning	0.5	Manual	Metric-based	Inductive
2022	MSSN* [34]	Contrastive Learning	1	Manual	Metric-based	Inductive
2022	EASY [35]	Rotation, Manifold Mixup	1	Manual	Transfer Learning-based	Inductive/Transductive
2023	Ji *et al.*$_{(GRSL)}$* [22]	Rotation, Contrastive Learning	1	Manual	Transfer Learning-based	Transductive
2023	CS^{2}TFSL* [36]	Rotation, Contrastive Learning	0.5	Manual	Transfer Learning-based	Transductive
2023	MFGNet* [37]	Rotation, Scale	1	Manual	Metric-based	Inductive
2023	MES^{2}L-Net* [21]	Semi-Arbitrary	1 for GLCL Average for Others	Manual	Transfer Learning-based	Inductive/Transductive
	CSENet (Ours)*	Arbitrary	Automatically Optimal	Autonomous	Transfer Learning-based	Inductive/Transductive

However, since the novel dataset does not contain the categories from the base dataset, it may lead to the collapse of the transferred model in the unseen categories. Therefore, the transfer learning paradigm relies on the excellent feature embedding [19].

Self-supervised learning is a promising solution aimed at improving the potential feature representation ability of few-shot models by defining auxiliary tasks to construct supervised information [25]. We summarized the current mainstream self-supervised few-shot methods as shown in Table I. Without loss of generality, current self-supervised few-shot classification approaches focus on supervising the model to learn more discriminative representations by constructing pseudo-labeling information with rotation, jigsaw puzzle, scales, and especially contrastive learning auxiliary tasks, to improve the generalized embedding for downstream few-shot classification. Nevertheless, the self-supervised auxiliary task (SSAT) weights in most current relevant algorithms are usually set manually or determined experimentally. It is worth noting that manual settings are usually empirical and difficult to quantitatively measure the contribution of the auxiliary task to the few-shot main task. Moreover, the experimental determinations for remote sensing datasets with different characteristics are costly and do not cover all combinations of true values. More importantly, most approaches with multiple pretext tasks assume that the auxiliary tasks are equally effective on the few-shot main task and search for the solutions that perform well on all tasks.

However, in realistic remote sensing scenarios, different scene classification datasets are different in terms of capture sensors, spatial resolutions, image sizes, scene categories, and data scales. These variations put the adaptive requirements on SSAT combinations. We take the four common SSATs of global contrastive learning (GCL) [38], spatial contrastive learning (SCL) [39], rotation prediction (RP) [40] and manifold mixup (MM) [19] as examples, and give the corresponding linear combination experiments on WHU-RS19 and UC Merced datasets as shown in Table II. Overall, it can be summarized in three critical problems for the current self-supervised few-shot scene classification algorithms.

TABLE II
ACCURACY COMPARISON OF DIFFERENT SSAT COMBINATIONS IN FIVE-WAY ONE-SHOT (INDUCTIVE INFERENCE)

Self-Supervised Auxiliary Task	5-way 1-shot	
	WHU-RS19	UC Merced
+GCL	**83.20**	56.11
+SCL	82.73	**58.63**
+RP	58.20	56.84
+MM	73.88	55.62
+GCL+SCL	**86.00**	58.64
+GCL+RP	72.00	51.81
+GCL+MM	85.26	**62.46**
+SCL+RP	75.02	52.98
+SCL+MM	84.96	58.22
+RP+MM	65.36	54.83
+GCL+SCL+RP	72.63	50.96
+GCL+SCL+MM	79.14	53.62
+GCL+RP+MM	**83.67**	**64.21**
+SCL+RP+MM	83.33	54.36
+GCL+SCL+RP+MM	**87.21**	**62.16**

1) The optimal self-supervised task linear combinations are not the same for different datasets, e.g., the optimal combinations on WHU-RS19 and UC Merced are "+GCL+SCL+RP+MM" and "+GCL+MM," respectively. Therefore, a personalized adaptive self-supervised combination is required for different remote sensing scene classification datasets.

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2) Uncertainty can exist in the performance improvement of SSAT combinations. Both complementary enhancement and mutual resistance to few-shot performance can exist when two or more pretext tasks are combined by simple linearity. Consequently, the intrinsic relationships between different auxiliary tasks are complex and do not lend themselves directly to simple linear combinations.
3) The above results only show the case of linear combinations and do not consider combinations with different weighting ratios. Intelligent adaptive weights with global optimization are called for better design in self-supervised few-shot scene classification.

To address the above challenges, in this article, we propose a collaborative self-supervised evolution framework, called CSENet. Instead of manually selecting the pretext tasks or setting the corresponding parameters according to the main task, CSENet can automatically search for the collaboration of SSATs and obtain the optimal combination of weights with the collaborative evolution strategy. The proposed framework not only reduces the dependence on manual parameter adjustment but also improves the generalization of the framework and the performance on the downstream few-shot classification task. The contributions of our work are summarized below.

1) A generalized CSENet framework is proposed, which can be adapted to arbitrary SSATs and automatically explore the optimal solution in different scenarios. We innovatively transform the self-supervised combination optimization problem into an evolutionary optimization problem, and CSENet supports both inductive and transductive inference based on the transfer learning training paradigm.
2) An automatic collaborative evolution strategy is designed to automatically search for appropriate weights in balancing the SSAT conflicts. The self-supervised weights are encoded as chromosomal genes, and mutation operation is implemented in the collaborative evolutionary strategy for autonomous global optimal search.
3) The proposed CSENet experiments on three benchmark remote sensing few-shot scene classification datasets, and the results demonstrate that the CSENet achieves state-of-the-art (SOTA) with reasonable weight combinations of pretext tasks in both inductive and transductive inference.

The remainder of this article is organized into the following structure. Section II presents the related work on few-shot scene classification and self-supervised learning. The background of evolutionary optimization in remote sensing is also described. The proposed CSENet framework is elaborated in Section III. Experimental results of the proposed CSENet are shown in Section IV. Conclusions are drawn in Section V.

II. Related Work

A. Few-Shot Scene Classification

Few-shot learning aims to train models with few labeled samples. In general, the task of scene classification requires a large amount of labeled data. However, it may be difficult to obtain the data in practical applications and the cost of labeling is expensive, especially in the field of remote sensing [41]. Therefore, the research of scene classification methods based on few-shot learning is of great significance for the interpretation of remote sensing images. Existing few-shot scene classification methods can be categorized into two kinds of methods based on the training approach, namely, meta-learning-based methods and transfer learning-based methods.

Meta-learning-based methods allow the model to quickly acquire the ability of "learning to learn," and then adapt to new scene classification tasks. In recent years, many researchers have proposed different meta-learning-based frameworks for few-shot scene classification, which combine different learning strategies to improve classification accuracy and model generalization capability. In [14], a lifelong few-shot learning algorithm was proposed for scene recognition in high-resolution remote sensing images to solve the problem of sample scarcity and large differences between datasets. Li et al. [13] proposed a deep meta-metric learning method called RS-MetaNet. RS-MetaNet organizes training through meta-tasks and learns the metric space that can well classify remote sensing scenes to make the algorithm generalizable for testing tasks. Similarly based on metric learning is [42], which improves the accuracy of classification by learning the similarity between local descriptors. Chen et al. [43] proposed a multimanifold metric learning framework to reduce the impact of intraclass differences and high interclass similarity. In [44], an iterative distribution learning network (IDLN) was proposed to incorporate multiple modules in a cyclic iterative architecture. In [45], the parameterized linear classifier (PLC) is integrated into the meta-learning framework to learn abundant features. In order to reduce undesired noise and improve the representation of embedded features, Yuan et al. [46] applied graph neural networks to few-shot scene classification and proposed a graph-based embedding smoothing network (GES-Net). To fully utilize the validity of prototypes, a Siamese-prototype network (SPNet) [47] was proposed to improve the representation of prototypes and enhance the interaction of information. In SGMNet [48], each test remote sensing image is represented as a graph, which captures co-occurring objects by graph nodes and spatial correlations by graph edges. To address the problem of interclass similarity and intraclass differences in RS images, a new multiscale covariance metric network (MCMNet) [49] was proposed to help identify the classes of unseen images by learning the feature differences from a few samples on multiple scales. Jia et al. [50] proposed an end-to-end framework containing a multiview integration (MI) module and a hard samples learning (HSL) module to explore hard samples in multiple views. In order to learn discriminative representations of remote sensing images, Li et al. [51] proposed a discriminative learning of adaptive match network (DLA-MatchNet). The model uses the incorporation of a channel attention module and spatial attention module to realize "discriminative learning." In [52], a new algorithm based on dictionary learning improves the feature discrimination ability and solves the "negative transfer" problem by embedding labeling information.

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Furthermore, a novel hybrid contrastive prototypical network was proposed in [17], which learns query-specific prototype representation with a contrastive learning framework and prototypical regularization to enhance interclass separation.

In order to improve the performance of the algorithm in a few labeled samples, the transfer learning-based methods transfer the model trained on large-scale datasets on a few-shot scene classification task. Ji et al. [20] proposed a framework for few-shot scene classification by utilizing calibrated prior tasks, which enhances the generalization ability of the model. In [19], the MM regularization technique was proposed to be robust to small differences in data distribution and enhance performance. Gong et al. [23] proposed a novel dual path aggregation attention network (DANet) combined with a quad-patch data augmentation strategy for solving the few-shot scene classification. In [22], a two-stage framework was proposed for few-shot remote sensing scene classification. The algorithm learns generic representations through auxiliary objectives in the first stage and propagates the knowledge through transductive inference in the second stage, which significantly improves the performance. In addition, Ji et al. [53] proposed a dual contrast network based on transfer learning to improve classification performance through contrast learning of context and detail information. In order to solve the problem of high interclass similarity and intraclass diversity in remote sensing images, Zhang et al. [54] proposed a frequency distribution-based multimodal fine-tuning strategy to alleviate the generalization limitations encountered in few-shot scenarios through the use of multimodal foundational models infused with basic domain knowledge.

B. Self-Supervised Learning

Self-supervised learning [55] is one of the unsupervised learning methods that does not require manual labels for training but automatically generates pseudo-labels on the input data for autonomous learning. Since the ability of self-supervised learning that utilizes a large amount of unlabeled data can alleviate the difficulty of label acquisition, it is widely used in various fields [56], [57], [58]. Pretext tasks in self-supervised learning can guide the model to learn useful feature representations. The pretext tasks are not the final tasks to be solved but are auxiliary tasks for the purpose of learning features. After training pretext tasks, the learned parameters are used as the pretrained model and the knowledge is transferred to other downstream tasks via fine-tuning.

Existing self-supervised learning methods can be categorized into generative, predictive, comparative, and noncomparative methods [59], which learn the features through different methods and are all applied to scene classification. Generative methods generally learn important features through a reconstruction process of the data. Lin et al. [60] proposed an innovative generative self-supervised learning method to solve the problem of scarce labeled data in remote sensing scene classification by utilizing the multiple-layer feature-matching generative adversarial network. In [61], a self-supervised learning network based on the encoder–decoder structure is proposed to learn image features and enhance the learning ability of the algorithm by considering location information with the image patch recovery strategy. In predictive methods, the model generally predicts the transformed data, which requires learning features related to semantic objects and regions. Yuan and Lin [62] proposed a transformer-based network to learn generic spectral–temporal representations related to land cover semantics, which can be transferred to specific classification tasks to improve the generalization ability of the model and reduce the risk of overfitting. Contrastive methods learn feature representations by comparing different perspectives of the same sample or similarities between different samples. For remote sensing images with spatio-temporal heterogeneity, Huang et al. [63] proposed a spatio-temporal invariant contrastive learning (STICL) framework to learn spatio-temporal invariant representations from unlabeled images that contain a large number of spatio-temporal scenes. The noncontrastive method is one type of joint embedding learning method which do not utilize contrastive components. For example, a gated self-attention module was introduced into GANs in [64] to focus on key scene regions and filter useless information, which enhances the feature representation. Jain et al. [65] proposed the RS-BYOL model, which is based on the BYOL [25] architecture and utilizes multispectral and SAR data as implicit augmentation to learn invariant feature embeddings.

The way of self-supervised learning utilizing unlabeled data for pretraining also provides new ideas for few-shot remote sensing scene classification. The model can be trained with reasonable pretext tasks in the pretraining stage, which enables the model to learn useful feature representations from unlabeled remote sensing data and improve its performance on downstream classification tasks. With pretraining, the model can achieve an improved performance on a limited number of labeled samples. In [31], self-attention and mutual-attention modules are applied in a task-specific contrastive learning method to learn feature correlation and reduce background interference. Li et al. [32] integrated multitask learning with self-supervised contrastive learning into a few-shot scene classification algorithm to take advantage of limited labeled data.

C. Evolutionary Optimization in Remote Sensing

Evolutionary algorithms (EAs) mimic the mechanisms of selection, inheritance, crossover, and mutation in the natural world, which search for optimal solutions in an iterative manner. An EA is able to globally search for a comprehensive solution in the feasible space, which is unlikely to fall into local optima. In addition, due to the fact that the evolutionary mechanism of EAs is easy to understand and implement while the parameters and strategies can be flexibly adapted to different tasks, EAs have been widely applied to various fields of remote sensing, such as sparse unmixing [66], SAR image reconstruction [67], neural architecture search for scene classification [68], and change detection [69].

Remote sensing data is characterized by various features such as high dimension and complex structure. EAs are able to search for feasible solutions in such data due to the powerful ability of knowledge discovery [70]. Therefore, many researchers have explored EAs for optimizing deep learning model architectures, which have shown excellent advantages

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in deep learning such as global search capability and wide adaptability, as well as achieving many good results. For example, in [68], a multiobjective neural evolution framework (SceneNet) was proposed for remote sensing scene classification. SceneNet optimizes the coded network structure through EAs, which achieves a more flexible hierarchical extraction of image information. Some other works tuned the model parameters through evolution strategies to help improve the overall accuracy. In addition, some researchers have used EAs to optimize different objectives in the original problem, which avoids setting the weighting parameters and obtains different tradeoff solutions. For example, Li et al. [66] considered reconstruction error and sparse error as two optimization objectives, and proposed evolutionary multitasking to explore the correlations between different sparse unmixing tasks.

III. Methodology

A. Problem Statement

In this article, the proposed CSENet is based on the transfer learning few-shot training paradigm. Therefore, the remote sensing scene classification datasets are divided into base dataset $\mathcal{D}_b$ and novel dataset $\mathcal{D}_n$ according to the execution phase of the algorithm. Among them, $\mathcal{D}_b$ contains traditional classified images with labels, which are independent of $\mathcal{D}_n$. The novel datasets are few-shot remote sensing images that can be further split into corresponding support sets and query sets.

To find the best combination of pretext tasks, we first perform a weight combination search with the automatic collaborative evolution strategy. In CSENet, the weights of different tasks are encoded as an individual in the population. Each individual is represented as $\boldsymbol{\omega}$. The number of pretext tasks corresponds to the length of chromosome U. The mathematical model of weight search can be expressed as

$$\min \mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \sum_{a=1}^{U} \omega^a \cdot \mathcal{L}_a \tag{1}$$

where $\mathcal{L}_{\mathrm{CE}}$ is the cross-entropy loss function of the main task and $\mathcal{L}_a$ is the loss function of ath pretext task. ω^a is the weight of $\mathcal{L}_a$ and also the ath position in the chromosome.

In order to reduce the complexity of the algorithm, the search phase only performs coarse learning. After searching for a suitable combination of weights, CSENet executes finer image feature learning. In the complete training phase, CSENet directly trains the model in the traditional image classification manner without dividing the training set into support and query sets. Instead, the novel dataset needs to be divided into support and query sets, where the support set contains images with labels and the query set contains without labels. In the testing phase, CSENet adopts the N-way K-shot setup. The original dataset contains N categories of images, among which K labeled images are randomly selected to form the support set. And images in the query set are all unlabeled and come from one of the N categories. After the training and testing phases, the proposed CSENet would not only find the best combination of weights for the pretext tasks but also obtain the optimal few-shot network.

B. Collaborative Self-Supervised Evolutionary Framework

CSENet consists of two phases, namely, the training phase and the testing phase.

1) Training: In the training phase, CSENet encodes distinct combinations of weights as individuals within a population and then searches for the optimal solution with an evolution strategy. In order to obtain good results, the proposed CSENet can adjust the evolution path in an adaptive manner based on historical information. In addition, in order to reduce the overall computational complexity of CSENet, the search process only learns the image features in a coarse way. The related evolution strategy will be described in detail in Section III-C.

The proposed CSENet adopts the transfer learning scheme for few-shot remote sensing scene classification. In the process of transfer learning, the training dataset does not contain any category in the few-shot remote sensing scene dataset, which increases the requirements for representation and generalization performance. Different from the existing transfer learning-based methods that directly pretrain the feature extractor with the training dataset, CSENet has coarse-learned features of the images in the search process. In addition, reasonable collaborative manners of pretext tasks are found to improve the generalization performance of the model. With the acquired feature learning foundation and the collaboration of multiple pretext tasks, CSENet is able to efficiently learn the detailed features of images.

2) Testing: In the testing phase, CSENet needs to apply the feature extractor obtained after training to the few-shot remote sensing scene. Since the training dataset is independent of the few-shot remote sensing scene dataset, the feature vector obtained from the feature extractor needs to be processed to help the CSENet for the subsequent classification. In order to help the classifier utilize the learned features effectively, the algorithm first centralizes the feature vectors and then projects them into the hypersphere, which can be specifically expressed as

$$\begin{aligned} \boldsymbol{v}_C &= \boldsymbol{v} - \bar{\boldsymbol{v}} \\ \boldsymbol{v}_H &= \frac{\boldsymbol{v}_C}{\|\boldsymbol{v}_C\|_2} \end{aligned} \tag{2}$$

where $\boldsymbol{v}$ is the feature vector extracted by the feature extractor. $\bar{\boldsymbol{v}}$ is calculated differently depending on the adopted classification method, i.e., the average feature vector of the base dataset (in inductive learning) or the average feature vector of the few-shot problems (in transductive learning).

Inductive learning and transductive learning are two existing mainstream approaches for few-shot remote sensing scene classification, which are both adopted in this article. The two approaches learn the image features with different data. In general, inductive learning is adopted in the case of data shortage, while transductive learning is more suitable for cases where the labels are lacking. Inductive learning takes a broader view of the data, which can learn both labeled and unlabeled image features and some additional information such as distributional clustering from their feature distributions to benefit the effectiveness. The nearest class mean (NCM) classifier is employed for class prediction in inductive learning, which focuses on finding the features that are closest to the

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barycenter for classification. The calculation is shown below

$$\bar{\boldsymbol{b}}_i = \frac{1}{|S_i|} \sum_{\boldsymbol{v}_H \in S_i} \boldsymbol{v}_H \tag{3}$$

where S_i is the support set of the ith class and $\boldsymbol{b}_i$ is the barycenter of this class. For any vector $\boldsymbol{v}_H$ in the query set Q, it can be categorized as follows:

$$C_I\left(\boldsymbol{v}_H, \left[\bar{\boldsymbol{b}}_1, \ldots, \bar{\boldsymbol{b}}_N\right]\right) = \arg\min_i \|\boldsymbol{v}_H - \bar{\boldsymbol{b}}_i\|_2. \tag{4}$$

In contrast to inductive learning, transductive learning learns general rules from the query set and applies them to the test images. Transductive learning enables learning from specific tasks to general tasks. With the lack of labeled data, transductive learning requires the generalization ability of the model, which is often practical in cases of difficult data labeling. The soft K-means (SKM) classifier is employed for class prediction in transductive learning, which predicts the class with constantly updated barycenters. The calculation of the barycenters can be written as

$$\begin{aligned} \bar{\boldsymbol{b}}_i^0 &= \frac{1}{|S_i|} \sum_{\boldsymbol{t}_H \in S_i} \boldsymbol{t}_H \\ \bar{\boldsymbol{b}}_i^{t+1} &= \sum_{\boldsymbol{t}_H \in S_i \cup Q} \frac{\eta\left(\boldsymbol{t}_H, \bar{\boldsymbol{b}}_i^t\right)}{\sum_{\boldsymbol{t}'_H \in S_i \cup Q} \eta\left(\boldsymbol{t}'_H, \bar{\boldsymbol{b}}_i^t\right)} \boldsymbol{t}_H \end{aligned} \tag{5}$$

where $\eta(\boldsymbol{t}_H, \bar{\boldsymbol{b}}_i^t)$ is the weight function of $\boldsymbol{t}_H$ and represents the probability of belonging to different classes. The calculation can be expressed as

$$\eta\left(\boldsymbol{t}_H, \bar{\boldsymbol{b}}_i^t\right) = \begin{cases} \dfrac{\exp\left(-\gamma \|\boldsymbol{t}_H - \bar{\boldsymbol{b}}_i^t\|_2^2\right)}{\sum_{j=1}^{N} \exp\left(-\gamma \|\boldsymbol{t}_H - \bar{\boldsymbol{b}}_j^t\|_2^2\right)}, & \text{if } \boldsymbol{t}_H \in Q \\ 1, & \text{if } \boldsymbol{t}_H \in S_i \end{cases} \tag{6}$$

where γ is the temperature value. In order to make the dataset and backbone results consistent, we take the value of γ to be 5 in the experiments. The classification prediction can be performed with the following formula:

$$C_T\left(\boldsymbol{t}_H, \left[\bar{\boldsymbol{b}}'_1, \ldots, \bar{\boldsymbol{b}}'_N\right]\right) = \arg\min_i \|\boldsymbol{t}_H - \bar{\boldsymbol{b}}'_i\|_2 \tag{7}$$

where $\bar{\boldsymbol{b}}'_i$ denotes the final barycenter.

C. Automatic Collaborative Evolution Strategy

The automatic collaborative evolution strategy for searching the weights of different tasks is shown in Algorithm 1. For the selection of auxiliary tasks, CSENet does not need to consider the characteristics of the main task or conduct experiments to determine whether the selected tasks are helpful to the main task. It only needs to enhance the completeness of the task set according to the limitation of computational resources, which means selecting the maximum number of auxiliary tasks aimed at enhancing different abilities as much as possible. In CSENet, $\boldsymbol{\omega}$ is assumed to be a solution for the optimization problem (1) and its probability distribution is $P_\theta(\boldsymbol{\omega})$. With the use of the evolution strategy to optimize the problem, the original problem is transformed into finding the parameter set θ of the optimal probability distribution $P_\theta(\boldsymbol{\omega})$.

Algorithm 1 Automatic Collaborative Evolution Strategy

Input: E: the population size; m: the number of parents; *MaxGen*: maximum number of generation; Θ: the set with U self-supervised auxiliary tasks.

Output: The weights ω for different tasks, the optimal few-shot network.

```
%Training
Initialize E individuals in the population and related parameters.
Evaluate individual fitness in the initial population with coarse training.
Calculate the mean vector μ and standard deviation σ of the initial population according to formula (10).
Set the current generation g ← 0.
while g ≤ MaxGen do
    Select m individuals as parents and generate o new offspring according to formula (11).
    Evaluating offspring fitness with coarse training.
    Obtaining the integrated population consisting of the original population and offspring.
    Selection of the top E individuals into the next generation.
    Record the best performing individual into set B.
    g ← g+1.
end while
```

The proposed method tracks the dependence between individuals through the covariance matrix $\boldsymbol{C}$ and consequently influences the generation of offspring. The parameter set θ and the probability distribution $P_\theta(\boldsymbol{\omega})$ can be expressed as

$$\begin{aligned} \theta &= (\boldsymbol{\mu}, \sigma, \boldsymbol{C}) \\ P_\theta(\boldsymbol{\omega}) &\sim \mathcal{N}\left(\boldsymbol{\mu}, \sigma^2 \boldsymbol{C}\right) \sim \boldsymbol{\mu} + \sigma \mathcal{N}(\boldsymbol{0}, \boldsymbol{C}) \end{aligned} \tag{8}$$

where $\boldsymbol{\mu}$ is the mean and σ is the standard deviation of $\boldsymbol{\omega}$.

1) Representation and Initialization: In order to find the optimal combination for the pretext tasks, CSENet encodes the weights of the loss functions as the individual $\boldsymbol{\omega}$ and adaptively explores the collaborative relationship between pretext tasks through evolutionary optimization. As shown in Fig. 1, the weight $\boldsymbol{\omega}^a$ of ath pretext task corresponds to ath position of the chromosome in an individual. Various individuals in the population represent different weights, which also imply different modes of collaboration for the pretext tasks.

In order to enable CSENet to efficiently determine the beneficial combinations from the pretext task set, we propose a new approach for individual initialization. Instead of generating initial individuals completely random, the proposed CSENet considers the combination with some absent tasks as individuals. When the weight of a loss function is 0, the corresponding task does not participate in self-supervised learning. CSENet realizes different collaboration of tasks by producing individuals with a weight of 0 in some tasks, which expands the diversity of the population. The distribution of the initialized population is shown in Fig. 2 and its computation

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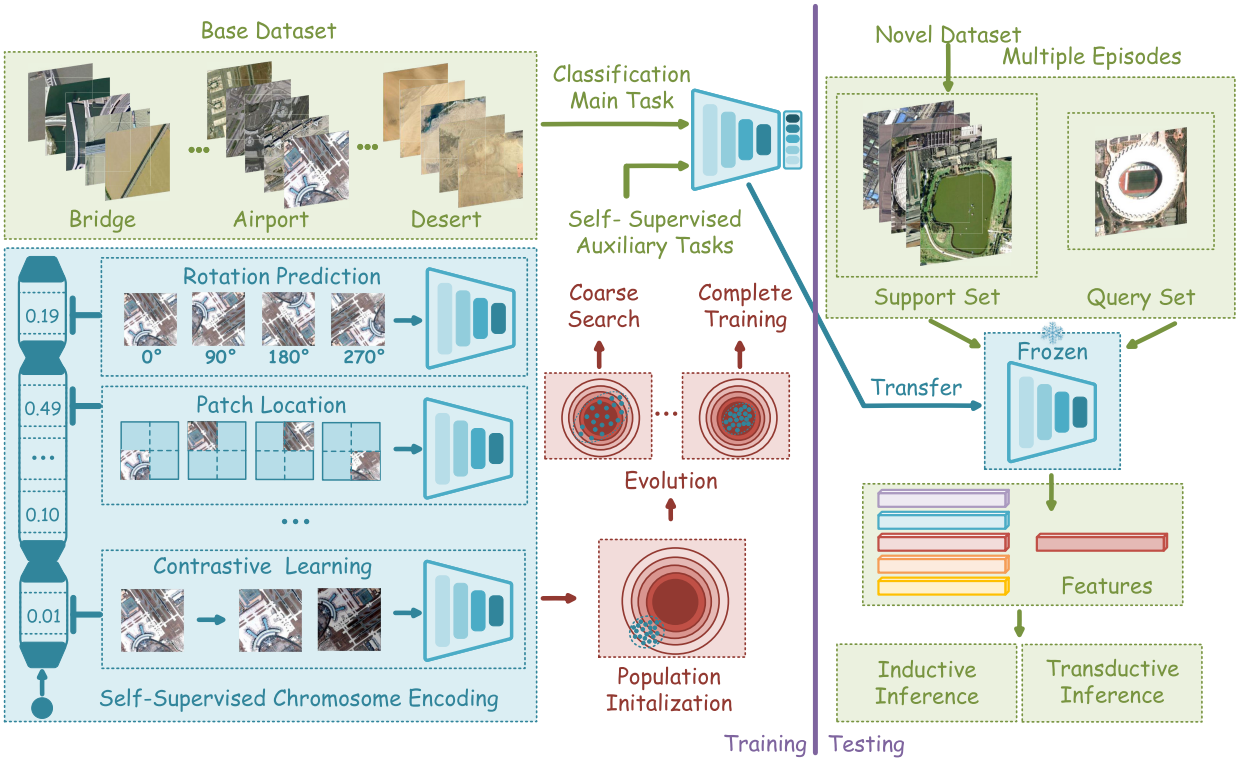


Fig. 1. Collaborative self-supervised evolutionary framework. In the training phase, we encode the weights of different tasks in the task set and train multiple classifiers through the coarse learning. After searching for the appropriate combination of weights, CSENet performs the complete training. In the testing phase, the trained feature extractor is frozen to process samples from support or query sets. The proposed framework supports both inductive inference and transductive inference for the classification of samples.

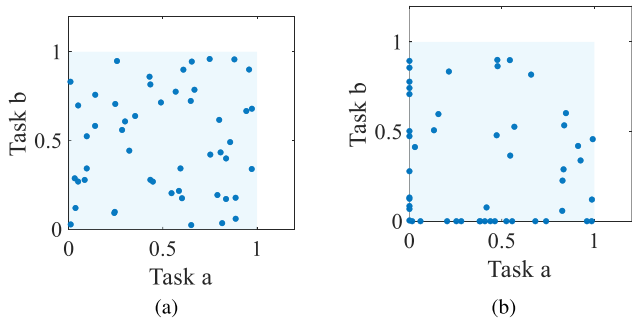


Fig. 2. Population distribution for different initialization methods. (a) Completely random initialization. (b) Multicombination initialization.

can be expressed as follows:

$$\omega^a = \begin{cases} 0, & r < \delta \\ \text{rand}(0,1), & \delta < r \leq 1 \end{cases} \tag{9}$$

where r is the factor to control the initialization, $r = \text{rand}(0, 1)$. δ is the multicombination initialization threshold which is set to 0.3 in the experiments. According to Fig. 2, it can be seen that most of the individuals initialized completely at random are distributed in scenarios where all tasks are considered. However, not all tasks have a positive effect on the overall performance. Using the proposed initialization method can obtain the distribution under different combinations of pretext tasks, which helps the proposed CSENet to find the global optimal solution by enhancing the diversity of individuals.

According to the initial population, the mean vector and standard deviation of the population can be further calculated as follows:

$$\boldsymbol{\mu}^{(0)} = \frac{1}{n}\sum_{i=1}^{n}\boldsymbol{\omega}_i^{(0)}$$
$$\sigma^{(0)} = \sqrt{\frac{1}{(n-1)}\sum_{i=1}^{n}\left(\boldsymbol{\omega}_i^{(0)} - \boldsymbol{\mu}^{(0)}\right)^2}. \tag{10}$$

2) Mutation: The proposed CSENet generates new solutions with the mutation strategy, which takes into account the information of the covariance matrix $\boldsymbol{C}$, mean $\boldsymbol{\mu}$ and standard deviation σ. The offspring can be sampled from the Gaussian distribution, which is expressed as follows:

$$\boldsymbol{O}^{(g+1)} = \left\{\boldsymbol{\omega}_i^{(g+1)} \mid \boldsymbol{\omega}_i^{(g+1)} = \boldsymbol{\mu}^{(g)} + \sigma^{(g)}\boldsymbol{y}_i^{(g+1)}, i = 1, \ldots, o\right\}$$
$$\boldsymbol{y}_i^{(g+1)} = \frac{\boldsymbol{\omega}_i^{(g+1)} - \boldsymbol{\mu}^{(g)}}{\sigma^{(g)}} \sim \mathcal{N}(\boldsymbol{\theta}, \boldsymbol{C}). \tag{11}$$

The covariance matrix $\boldsymbol{C}$ keeps track of the dependence between individuals, which enables CSENet to quickly adjust the exploration space as the confidence changes. The standard deviation σ can be used as the step size, which determines the extent of the exploration. A larger step size indicates a wider search space. The mean value $\boldsymbol{\mu}$ represents the center of each search. Therefore, the offspring are generated according to the performance of the current population and their dependencies on each other, which can help the algorithm to search in an efficient manner.

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3) Automatic Collaborative Evolution: In this article, a CSENet is proposed to search for a way to collaborate different pretext tasks in self-supervised learning. In the evolution process, the parameters represented by the individual are input into the model for training, and then the testing result is used as the individual fitness value. In order to balance the search efficiency and accuracy, CSENet only performs coarse learning of image features in this phase, as shown in Algorithm 1. The fitness of each individual measures the performance of the weight combination in the main task. For task combinations with negative effects, the corresponding individuals will also perform poorly in terms of fitness. Such individuals will be gradually eliminated in the evolutionary process, and tasks with positive effects on the main task will be left to further explore their optimal collaborative weights.

After the evaluation of the individual fitness, CSENet updates the corresponding parameters according to the performance of the individuals in the population. The mean $\boldsymbol{\mu}$ determines the center of each search, which is updated as

$$\boldsymbol{\mu}^{(g+1)} = \boldsymbol{\mu}^{(g)} + \alpha_\mu \frac{1}{m} \sum_{i=1}^{m} \left(\boldsymbol{\omega}_i^{(g+1)} - \boldsymbol{\mu}^{(g)} \right) \tag{12}$$

where α_μ is a learning rate, usually set to 1 and m is the number of parents. The parameter σ controls the overall scale of the distribution. CSENet is able to adaptively adjust the step size σ, which is updated as follows:

$$\begin{aligned} \sigma^{(g+1)} &= \sigma^{(g)} \exp\left(\frac{\alpha_\sigma}{d_\sigma} \left(\frac{\|\boldsymbol{p}_\sigma^{(g+1)}\|}{\mathbb{E}\|\mathcal{N}(\boldsymbol{0}, \boldsymbol{I})\|} - 1 \right) \right) \\ \boldsymbol{p}_\sigma^{(g+1)} &= (1-\alpha_\sigma)\boldsymbol{p}_\sigma^{(g)} + \sqrt{\alpha_\sigma(2-\alpha_\sigma)m_e}\boldsymbol{C}^{(g)^{-\frac{1}{2}}} \frac{\boldsymbol{\mu}^{(g+1)} - \boldsymbol{\mu}^{(g)}}{\sigma^{(g)}} \end{aligned} \tag{13}$$

where $\boldsymbol{p}_\sigma$ is the conjugate evolution path. α_σ is the backward time horizon of the evolution path and d_σ is the damping parameter.

For the estimation of the covariance matrix, the combination of rank-m and rank-one is used to generate a stable and reliable covariance matrix $\boldsymbol{C}$. The specific calculation is given as follows:

$$\begin{aligned} \boldsymbol{C}^{(g+1)} = &(1 - c_1 - c_m)\boldsymbol{C}^{(g)} \\ &+ c_1 \underbrace{\boldsymbol{p}_{\mathrm{c}}^{(g+1)} \boldsymbol{p}_{\mathrm{c}}^{(g+1)^\top}}_{\text{rank-one update}} + c_m \underbrace{\frac{1}{e} \sum_{i=1}^{e} \boldsymbol{y}_i^{(g+1)} \left(\boldsymbol{y}_i^{(g+1)} \right)^\top}_{\text{rank-m update}} \end{aligned} \tag{14}$$

$$\boldsymbol{p}_{\mathrm{c}}^{(g+1)} = (1-\alpha_{\mathrm{c}})\boldsymbol{p}_{\mathrm{c}}^{(g)} + \sqrt{\alpha_{\mathrm{c}}(2-\alpha_{\mathrm{c}})m_e} \frac{\boldsymbol{\mu}^{(g+1)} - \boldsymbol{\mu}^{(g)}}{\sigma^{(g)}} \tag{15}$$

where $\boldsymbol{p}_{\mathrm{c}}$ is the evolution path and α_{c} is the decay rate for cumulative path. c_1 and c_m are the rank-one update learning rate and the rank-m update learning rate, respectively. E is the population size. In CSENet, the covariance matrix can be adaptively adjusted to search for solutions more efficiently.

IV. Experimental Results

A. Datasets

1) WHU-RS19 Dataset: The WHU-RS19 dataset [73] is one of the widely used remote sensing image datasets for scene classification, which was released by Wuhan University in 2012. The dataset contains 950 images with a size of 600×600 pixels and is uniformly distributed in 19 typical classes of high-resolution satellite image scenes. Each class of scene includes about 50 samples. The samples were taken from different regions with different scales, orientations, and illuminations, which enables a comprehensive test on the algorithmic accuracy of the classification.

2) UC Merced Land-Use Dataset: The UC Merced Land-Use dataset [74] is from the USGS National Map Urban Area Imagery collection, which contains 2100 images with 256×256 pixels. Images captured from various regions across different cities which ultimately comprised 21 scene categories, including buildings, forests, freeways, and other classes. Each scene category comprises approximately 100 samples. This dataset is also widely utilized in the field of scene classification.

3) NWPU-RESISC45 Dataset: The NWPU-RESISC45 dataset [75] was released by Northwestern Polytechnical University. The dataset contains 31 500 images covering 45 categories. Each category includes 700 images of size 256×256 pixels. Due to its abundant scene categories and image information, NWPU-RESISC45 is widely used in scene classification. In addition, its large-scale image data can also help to test the classification accuracy of different algorithms.

B. Experimental Setup

The number of training epochs for the coarse search is set to 10, and CSENet adopts 500 as the final number of epochs for the complete training. The population size is set to 50. To balance the search speed and the performance effect, the number of evolution generations is set to 50. In the testing phase, two approaches (five-way one-shot and five-way five-shot) are used to observe the performance in different modes. In this article, the learning parameters for evolution are set in line with [76].

Four kinds of self-supervised pretext tasks are selected to form the task set in the experiment, i.e., GCL, SCL, RP, and MM. Contrastive learning is one of the common self-supervised pretext tasks for distinguishing samples from two different distributions. GCL focuses on global image features and typically uses different pixels in an image or similarities between images to construct contrastive loss. It is able to learn the global semantic information in the image, and the features are learned more comprehensively, which is suitable for tasks that require global information. SCL, on the other hand, is able to learn detailed information about the local region in the image and is suitable for tasks that need to focus on the local structure of the image. The selection of GCL and SCL as pretext tasks in the experiments not only enables the analysis of their differences in the scene classification task but also explores the collaboration between the two tasks. RP is an effective self-supervised pretext task, which increases the diversity of the data by rotating images and enables the model to comprehensively learn and understand the features at different angles and orientations. In addition, RP helps to alleviate the overfitting problem and reduces the dependence of the model on specific angles, which improves

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TABLE III
OVERALL ACCURACY (IN %) COMPARISON OF DIFFERENT FEW-SHOT REMOTE SENSING SCENE CLASSIFICATION METHODS ON THE WHU-RS19 DATASET

Method	5-way	
	1-shot	5-shot
Meta-learning-based Methods		
MatchingNet [8]	60.14 ± 0.67	68.66 ± 0.38
MAML [9]	49.13 ± 0.65	62.49 ± 0.51
ProtoNet [10]	69.03 ± 0.65	84.84 ± 0.33
TPN† [11]	66.51 ± 0.87	78.50 ± 0.56
Relation Net [12]	70.01 ± 0.68	86.04 ± 0.32
LLSR [14]	57.10	70.65
DLA-MatchNet [51]	68.27 ± 1.83	79.89 ± 0.33
IDLN [44]	73.89 ± 0.88	83.12 ± 0.56
PCFGNet [71]	72.41 ± 0.91	85.26 ± 0.66
GES-Net† [46]	75.84 ± 0.78	82.37 ± 0.38
TAE-Net [72]	73.67 ± 0.74	88.95 ± 0.52
SPNet [47]	81.06 ± 0.60	88.04 ± 0.28
TSC [31]	70.99 ± 0.74	82.18 ± 0.32
SGMNet [48]	85.06 ± 0.55	90.12 ± 0.27
Ma *et al.* [52]	82.57 ± 0.58	93.33 ± 0.23
MLLD [42]	76.07 ± 0.65	90.69 ± 0.27
MCMNet [49]	72.73 ± 0.48	84.54 ± 0.23
HSL-MINet [50]	87.33 ± 0.37	91.22 ± 0.14
MMML [43]	77.76 ± 0.69	88.40 ± 0.54
Transfer learning-based Methods		
baseline [18]	75.57 ± 0.36	88.65 ± 0.18
S2M2 [19]	69.00 ± 0.41	82.14 ± 0.21
Ji *et al.*(TGRS) [20]	86.29 ± 0.30	92.96 ± 0.12
Ji *et al.*(GRSL) [22]	88.00 ± 0.29	94.17 ± 0.12
Ji *et al.*†(GRSL) [22]	91.58 ± 0.28	94.24 ± 0.12
DANet [23]	75.02 ± 0.16	89.21 ± 0.07
MES^{2}L-Net [21]	87.21 ± 0.12	93.58 ± 0.07
MES^{2}L-Net† [21]	$\underline{92.21 \pm 0.05}$	$\underline{95.03 \pm 0.04}$
CSENet	89.16 ± 0.12	95.62 ± 0.05
CSENet†	$\mathbf{95.53 \pm 0.10}$	$\mathbf{97.43 \pm 0.03}$

TABLE IV
OVERALL ACCURACY (IN %) COMPARISON OF DIFFERENT FEW-SHOT REMOTE SENSING SCENE CLASSIFICATION METHODS ON THE UC MERCED LAND-USE DATASET

Method	5-way	
	1-shot	5-shot
Meta-learning-based Methods		
MatchingNet [8]	46.46 ± 0.75	67.39 ± 0.50
MAML [9]	48.86 ± 0.74	60.78 ± 0.62
ProtoNet [10]	54.03 ± 0.78	70.81 ± 0.48
TPN† [11]	53.36 ± 0.77	68.23 ± 0.52
Relation Net [12]	49.77 ± 0.78	57.98 ± 0.57
LLSR [14]	39.47	57.40
RS-MetaNet [13]	57.23 ± 0.56	76.08 ± 0.28
DLA-MatchNet [51]	53.76 ± 0.62	63.01 ± 0.51
MKN [45]	58.45 ± 0.54	77.92 ± 0.32
SCL-MLNet [32]	51.37 ± 0.79	68.09 ± 0.92
GES-Net† [46]	58.88 ± 0.81	81.66 ± 0.50
TAE-Net [72]	60.21 ± 0.72	77.44 ± 0.51
SPNet [47]	57.64 ± 0.73	73.52 ± 0.51
TSC [31]	55.11 ± 0.68	69.20 ± 0.64
SGMNet [48]	60.52 ± 0.74	73.42 ± 0.49
Ma *et al.* [52]	65.44 ± 0.72	82.63 ± 0.45
MLLD [42]	59.05 ± 0.75	77.76 ± 0.52
MCMNet [49]	56.58 ± 0.39	71.35 ± 0.46
HSL-MINet [50]	66.32 ± 0.11	79.83 ± 0.14
MMML [43]	61.42 ± 0.34	79.95 ± 0.16
Transfer learning-based Methods		
baseline [18]	60.41 ± 0.38	77.02 ± 0.24
S2M2 [19]	56.42 ± 0.40	71.97 ± 0.27
DANet [23]	61.60 ± 0.18	78.02 ± 0.16
MES^{2}L-Net [21]	64.21 ± 0.17	80.95 ± 0.15
MES^{2}L-Net† [21]	$\underline{70.73 \pm 0.16}$	$\underline{82.92 \pm 0.12}$
CSENet	66.16 ± 0.18	81.31 ± 0.11
CSENet†	$\mathbf{72.60 \pm 0.26}$	$\mathbf{84.05 \pm 0.11}$

the generalization performance. MM is a method based on data augmentation and deep learning regularization, which can effectively explore the potential structure of the data by interpolating within the feature space and improving the generalization ability and robustness of the model. In remote sensing scene classification, MM can generate new image samples by mixing in the representation space of remote sensing images, which enriches the training dataset and increases the diversity of data.

C. Comparison With SOTA for Few-Shot Scene Classification

In order to validate the effectiveness of the proposed method, we adopted a variety of few-shot scene classification methods as comparisons and obtained the results on different datasets. The comparison methods are all from recent few-shot scene classification studies, which design different strategies to help the algorithms as well as take different inference approaches. Based on the training approach, these methods can be categorized into meta-learning-based methods (MatchingNet [8], MAML [9], ProtoNet [10], TPN [11], Relation Net [12], LLSR [14], DLA-MatchNet [51], RS-MetaNet [13], IDLN [44], MKN [45], SCL-MLNet [32], PCFGNet [71], GES-Net [46], TAE-Net [72], SPNet [47], TSC [31], SGMNet [48], Ma et al. [52], MLLD [42], MCMNet [49], HSL-MINet [50], and MMML [43]) and transfer learning-based methods (baseline [18], S2M2 [19], Ji et al. (TGRS) [20], Ji et al. (GRSL) [22], DANet [23], and MES^{2}L-Net [21]). Among all the methods, MES^{2}L-Net [21] and the proposed CSENet support both inductive and transductive inferences. In the remaining methods, Ji et al. (GRSL) [22], TPN [11], and GES-Net [46] are transductive inference and all other methods are inductive inference.

Since the proposed methods are based on transfer learning, we group the different classes of methods in the results (meta-learning-based methods, transfer learning-based methods) for a clear comparison. The results of all the methods on different datasets are represented in Tables III–V. The best results are denoted in bold and the second best results are underlined for representation. In addition, methods with the "†" flag represent the results of transductive learning and others represent the results of inductive learning.

Table III demonstrates the classification accuracy of different methods on the WHU-RS19 dataset. In the table, the results of transductive inference are generally better than those of inductive inference. For inductive inference, the classification results of CSENet on one-shot and five-shot are over the

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TABLE V
OVERALL ACCURACY (IN %) COMPARISON OF DIFFERENT FEW-SHOT REMOTE SENSING SCENE CLASSIFICATION METHODS ON THE NWPU-RESISC45 DATASET

Method	5-way	
	1-shot	5-shot
Meta-learning-based Methods		
MatchingNet [8]	45.80 ± 0.86	64.35 ± 0.65
MAML [9]	48.40 ± 0.82	62.90 ± 0.69
ProtoNet [10]	62.43 ± 0.84	78.68 ± 0.55
TPN† [11]	66.51 ± 0.87	78.50 ± 0.56
Relation Net [12]	66.58 ± 0.91	76.11 ± 0.58
RS-MetaNet [13]	52.78 ± 0.09	71.49 ± 0.81
LLSR [14]	51.43	72.90
DLA-MatchNet [51]	68.80 ± 0.70	81.63 ± 0.46
IDLN [44]	75.25 ± 0.75	84.67 ± 0.23
PCFGNet [71]	72.05 ± 0.75	85.07 ± 0.45
GES-Net† [46]	70.83 ± 0.85	82.27 ± 0.55
MKN [45]	65.84 ± 0.89	82.67 ± 0.55
SCL-MLNet [32]	62.21 ± 1.12	80.86 ± 0.76
TAE-Net [72]	69.13 ± 0.83	82.37 ± 0.52
SPNet [47]	67.84 ± 0.87	83.94 ± 0.50
TSC [31]	73.26 ± 0.15	84.62 ± 0.35
SGMNet [48]	70.40 ± 0.83	82.32 ± 0.53
Ma *et al.* [52]	74.03 ± 0.76	88.32 ± 0.43
MLLD [42]	65.88 ± 0.83	82.06 ± 0.53
MCMNet [49]	74.61 ± 0.18	86.57 ± 0.42
HSL-MINet [50]	76.71 ± 0.39	88.18 ± 0.23
MMML [43]	77.35 ± 0.73	88.19 ± 0.47
Transfer learning-based Methods		
baseline [18]	69.02 ± 0.46	85.62 ± 0.25
S2M2 [19]	63.24 ± 0.47	83.23 ± 0.28
Ji *et al.*(TGRS) [20]	76.70 ± 0.44	89.87 ± 0.21
Ji *et al.*(GRSL) [22]	73.50 ± 0.44	87.90 ± 0.21
Ji *et al.*†(GRSL) [22]	80.58 ± 0.48	89.20 ± 0.21
DANet [23]	74.30 ± 0.20	87.29 ± 0.11
MES^{2}L-Net [21]	79.37 ± 0.20	89.94 ± 0.11
MES^{2}L-Net† [21]	$\underline{86.55 \pm 0.18}$	$\underline{91.06 \pm 0.11}$
CSENet	81.28 ± 0.14	91.47 ± 0.09
CSENet†	$\mathbf{89.49 \pm 0.21}$	$\mathbf{92.87 \pm 0.14}$

second-ranked method by 1.16% and 1.45%, while the results of transductive inference are over 3.32% and 2.40%, respectively. In contrast to using all pretext tasks equally or setting weights manually, the collaboration of pretext tasks searched by the CSENet would be more effective in assisting the main task. Similar phenomena are observed for the UC Merced and NWPU-RESISC45 datasets, where CSENet obtains the best results for both inductive and transductive inference. The automatic collaborative evolution strategy is able to adaptively explore the implicit association information between pretext tasks and the main task, which can be used to search for the optimal collaborative solution to enhance the performance of the main task. In addition, this adaptive collaborative approach also improves the robustness of the algorithm in different data, which contributes to the fact that CSENet obtains excellent results in all three datasets.

D. Experiments on the CSENet

In the experiment, we selected four self-supervised pretext tasks to be randomly combined with the main task and

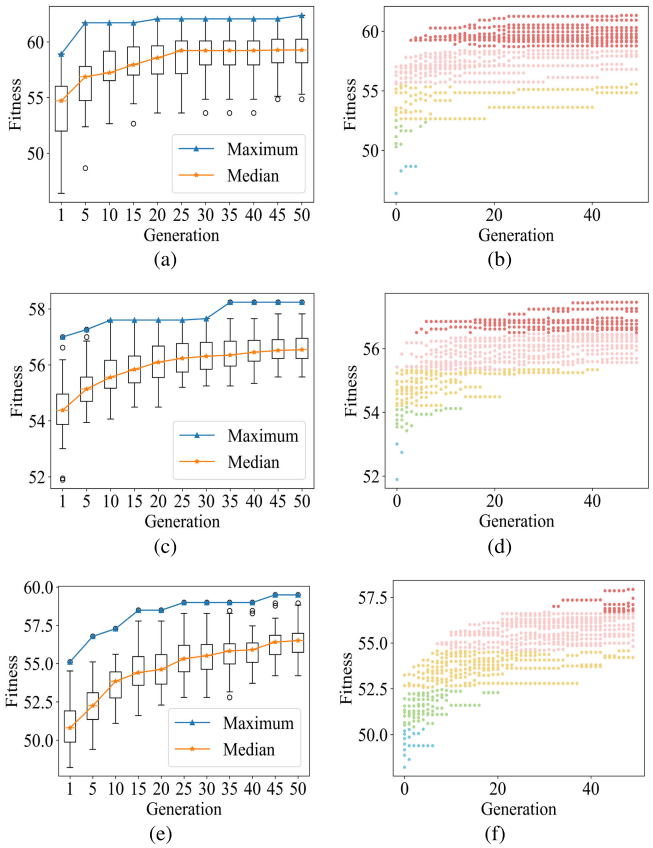


Fig. 3. Box plot of evolution and evolution of individual fitness. (a) and (b) Results on the WHU-RS19 dataset. (c) and (d) Results on the UC Merced dataset. (e) and (f) Results on the NWPU-RESISC45 dataset.

explored with possible combinations, which were performed on three datasets, respectively. The weights of the combinations are all set to 1, which means that each auxiliary task is assumed to be equally effective for the main task. The results of the experiments are shown in Table VI, where the "Baseline" indicates that no self-supervised learning task is used. It can be seen from Table VI that the combination of RP and MM on the WHU-RS19 dataset is less effective than MM alone. However, when RP is added to the combination of GCL, SCL, and MM, the performance of CSENet reaches an improvement. For the NWPU-RESISC45 dataset, the combination of GCL, SCL, RP, and MM works well for inductive inference, while the combination of SCL, RP, and MM performs better for transductive inference. These results confirm that different pretext tasks do not benefit the main task to the same extent, and some tasks may even have negative effects. In addition, the best combination of tasks changes on different datasets. However, the proposed method is able to explore the optimal collaborative combinations of different self-supervised learning tasks and avoid introducing negatively affected pretext tasks for training. In addition, CSENet can also adjust the weights of different pretext tasks according to their benefits to help improve the overall effect of the algorithm. The best accuracy of CSENet on different datasets demonstrates the importance of collaborative self-supervised learning.

In order to analyze the search process of the automatic collaborative evolution, the box plots of different generations

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TABLE VI
QUANTITATIVE ANALYSIS RESULTS (IN %) FOR DIFFERENT NUMBER OF SSATS ON THEIR REMOTE SENSING SCENE CLASSIFICATION DATASETS

Method	WHU-RS19		UC Merced		NWPU-RESISC45	
	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
Inductive Inference						
Baseline	65.36 ± 0.16	82.74 ± 0.09	50.99 ± 0.24	69.36 ± 0.17	65.64 ± 0.21	80.65 ± 0.13
+GCL	83.20 ± 0.13	92.48 ± 0.06	56.11 ± 0.18	73.14 ± 0.12	76.48 ± 0.20	88.40 ± 0.14
+SCL	82.73 ± 0.13	93.09 ± 0.06	58.63 ± 0.18	74.05 ± 0.12	51.64 ± 0.21	66.42 ± 0.17
+RP	58.20 ± 0.15	77.21 ± 0.11	56.84 ± 0.18	72.67 ± 0.13	77.60 ± 0.21	88.39 ± 0.15
+MM	73.88 ± 0.17	89.82 ± 0.08	55.62 ± 0.18	72.30 ± 0.13	75.00 ± 0.21	86.58 ± 0.15
+GCL+SCL	86.00 ± 0.12	91.57 ± 0.05	58.64 ± 0.19	75.12 ± 0.12	77.39 ± 0.20	88.92 ± 0.14
+GCL+RP	72.00 ± 0.18	88.86 ± 0.08	51.81 ± 0.17	70.72 ± 0.13	74.65 ± 0.22	87.80 ± 0.16
+GCL+MM	85.26 ± 0.13	93.30 ± 0.07	62.46 ± 0.18	79.69 ± 0.10	69.96 ± 0.20	85.66 ± 0.11
+SCL+RP	75.02 ± 0.17	91.42 ± 0.07	52.98 ± 0.18	69.15 ± 0.14	75.50 ± 0.21	87.61 ± 0.16
+SCL+MM	84.96 ± 0.14	91.37 ± 0.05	58.22 ± 0.18	77.05 ± 0.12	77.54 ± 0.20	88.62 ± 0.15
+RP+MM	65.36 ± 0.17	84.09 ± 0.10	54.83 ± 0.17	71.01 ± 0.13	76.83 ± 0.21	88.38 ± 0.15
+GCL+SCL+RP	72.63 ± 0.18	90.31 ± 0.07	50.96 ± 0.17	69.30 ± 0.13	76.91 ± 0.21	88.39 ± 0.16
+GCL+SCL+MM	79.14 ± 0.15	92.29 ± 0.07	53.62 ± 0.18	71.28 ± 0.13	78.60 ± 0.20	89.47 ± 0.14
+GCL+RP+MM	83.67 ± 0.14	92.28 ± 0.06	64.21 ± 0.18	80.95 ± 0.11	78.19 ± 0.20	88.96 ± 0.14
+SCL+RP+MM	83.33 ± 0.14	91.17 ± 0.05	54.36 ± 0.18	71.08 ± 0.13	79.09 ± 0.20	89.80 ± 0.13
+GCL+SCL+RP+MM	87.21 ± 0.12	93.58 ± 0.07	62.16 ± 0.19	78.13 ± 0.11	79.37 ± 0.20	89.94 ± 0.11
CSENet	$\mathbf{89.16 \pm 0.12}$ $\mathbf{(+1.95)}$	$\mathbf{95.62 \pm 0.05}$ $\mathbf{(+2.04)}$	$\mathbf{66.16 \pm 0.18}$ $\mathbf{(+1.95)}$	$\mathbf{81.31 \pm 0.11}$ $\mathbf{(+0.36)}$	$\mathbf{81.28 \pm 0.14}$ $\mathbf{(+1.91)}$	$\mathbf{91.47 \pm 0.09}$ $\mathbf{(+1.53)}$
Transductive Inference						
Baseline†	70.16 ± 0.21	84.30 ± 0.10	52.98 ± 0.24	74.01 ± 0.15	73.07 ± 0.29	81.90 ± 0.15
+GCL†	91.21 ± 0.14	93.19 ± 0.05	59.80 ± 0.27	73.05 ± 0.13	86.63 ± 0.21	89.83 ± 0.14
+SCL†	91.54 ± 0.15	94.28 ± 0.05	61.28 ± 0.27	72.82 ± 0.13	50.95 ± 0.28	63.39 ± 0.19
+RP†	70.16 ± 0.22	80.39 ± 0.10	63.09 ± 0.23	73.30 ± 0.14	86.99 ± 0.21	89.60 ± 0.15
+MM†	87.80 ± 0.20	91.83 ± 0.06	57.32 ± 0.27	71.87 ± 0.14	84.00 ± 0.23	87.77 ± 0.15
+GCL+SCL†	90.27 ± 0.13	94.85 ± 0.05	62.12 ± 0.27	74.29 ± 0.14	87.79 ± 0.20	90.34 ± 0.14
+GCL+RP†	83.64 ± 0.17	88.83 ± 0.06	58.28 ± 0.21	71.29 ± 0.12	84.10 ± 0.25	89.07 ± 0.16
+GCL+MM†	91.66 ± 0.10	92.99 ± 0.06	70.71 ± 0.27	80.88 ± 0.10	86.38 ± 0.22	89.71 ± 0.15
+SCL+RP†	86.03 ± 0.13	93.09 ± 0.05	56.38 ± 0.23	67.89 ± 0.15	83.67 ± 0.25	88.48 ± 0.17
+SCL+MM†	90.88 ± 0.12	94.03 ± 0.04	64.31 ± 0.28	78.48 ± 0.12	82.69 ± 0.23	88.14 ± 0.10
+RP+MM†	72.06 ± 0.26	84.28 ± 0.11	59.37 ± 0.25	71.31 ± 0.14	86.39 ± 0.22	89.69 ± 0.15
+GCL+SCL+RP†	85.76 ± 0.17	92.32 ± 0.06	52.99 ± 0.20	67.92 ± 0.13	86.15 ± 0.23	89.53 ± 0.16
+GCL+SCL+MM†	89.35 ± 0.17	93.43 ± 0.06	51.73 ± 0.25	67.87 ± 0.15	87.49 ± 0.21	90.47 ± 0.14
+GCL+RP+MM†	91.47 ± 0.10	94.23 ± 0.04	70.73 ± 0.26	82.95 ± 0.12	86.52 ± 0.22	89.97 ± 0.14
+SCL+RP+MM†	91.33 ± 0.13	93.90 ± 0.04	54.94 ± 0.25	68.55 ± 0.15	88.55 ± 0.20	91.06 ± 0.11
+GCL+SCL+RP+MM†	92.21 ± 0.05	95.03 ± 0.04	66.46 ± 0.28	79.40 ± 0.13	86.55 ± 0.18	91.06 ± 0.11
CSENet†	$\mathbf{95.53 \pm 0.10}$ $\mathbf{(+3.32)}$	$\mathbf{97.43 \pm 0.03}$ $\mathbf{(+2.40)}$	$\mathbf{72.60 \pm 0.26}$ $\mathbf{(+1.87)}$	$\mathbf{84.05 \pm 0.11}$ $\mathbf{(+1.10)}$	$\mathbf{89.49 \pm 0.21}$ $\mathbf{(+0.94)}$	$\mathbf{92.87 \pm 0.14}$ $\mathbf{(+1.81)}$

and individual fitness evolution on different datasets are shown in Fig. 3. In the individual fitness evolution, the darker point indicates the better adaptation of the corresponding individual. Ten individuals of the population are selected for display in each generation, including the individuals with the largest and the smallest fitness, and the remaining eight individuals are sampled uniformly in the population. Therefore, the figure reflects the density of individuals. The box plot of population evolution shows the distribution of fitness across the entire population, which is plotted at five-generation intervals.

It can be seen from Fig. 3 that the fitness of the entire population increases as the number of generations increases, and the narrowing of the box indicates that the distribution of the individuals is gradually stabilizing. Meanwhile, Fig. 3(d) also shows that the individuals in the population gradually become denser with the process of evolution. At the beginning, the fitness is scattered between 52% and 57%. In the end, most of the individuals have fitness between 55% and 57%. The strongest individual has increased from less than 57% to more than 58%, which indicates that the proposed CSENet is effective. For the WHU-RS19 dataset, CSENet shows good fitness after the fifth generation. It is worth noting that there are still some poorly adapted individuals in the final population, which is due to the fact that the automatic collaborative

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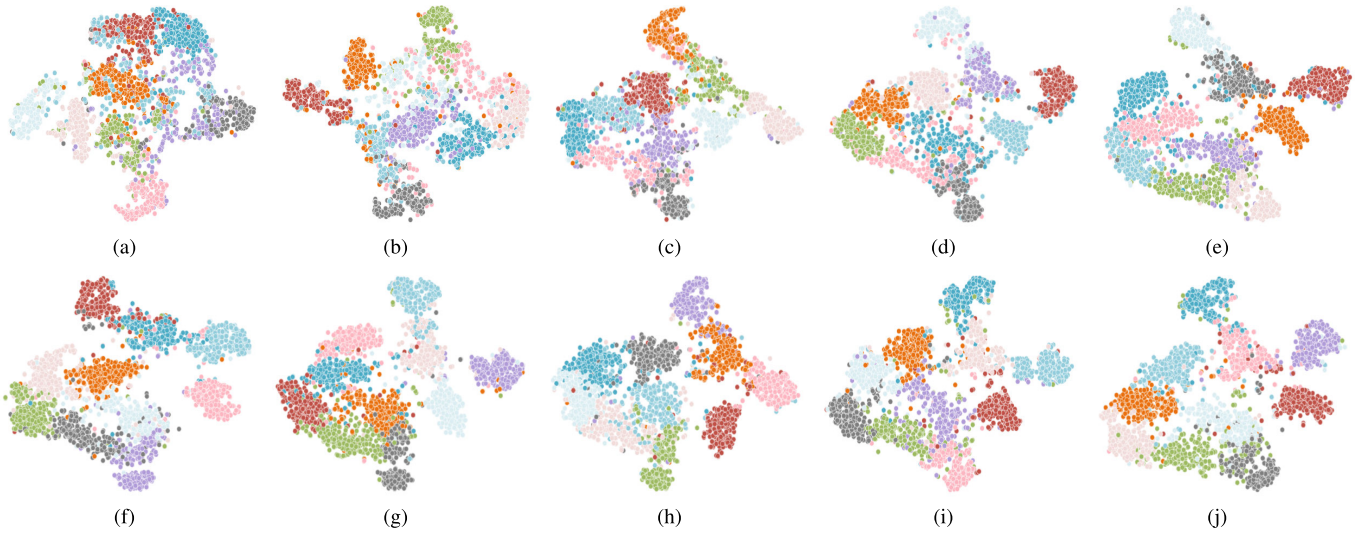


Fig. 4. Comparison of t-SNE visualization in different evolution generations. (a) Generation = 5. (b) Generation = 10. (c) Generation = 15. (d) Generation = 20. (e) Generation = 25. (f) Generation = 30. (g) Generation = 35. (h) Generation = 40. (i) Generation = 45. (j) Generation = 50.

evolution strategy is a global search strategy. Even after CSENet has converged toward an optimal solution, it continues to explore the solution space for potentially new and better individuals. Since the NWPU-RESISC45 dataset contains multiple scene categories and has a large number of samples, more generations are needed for convergence. However, the proposed CSENet is completely trained after inheriting the coarsely learned parameters, which improves the reliability of initial parameters and also allows CSENet to converge with a few epochs. The t-SNE visualization comparison on the NWPU-RESISC45 dataset is given in Fig. 4, which depicts the feature learned through the coarse-trained collaborative network at different evolutionary generations. It can be intuitively observed the similarities and differences between different categories. The collaborative effect of self-supervised tasks becomes increasingly evident as the number of evolutionary generations advances. In addition, the data points in various categories become more distinctly separated within the feature space as generation increases, which demonstrates that the proposed collaborative self-supervised evolutionary framework can help the algorithms to learn the feature representation.

In order to specifically analyze the help of the collaborative effect on the CSENet, the accuracy of CSENet with complete training on different datasets is shown in Fig. 5. In the figure, "Baseline" is the method without adding any pretext tasks. We have adopted the "Baseline" as the foundational framework, to which we have sequentially integrated a varying number of pretext tasks. The best results with different number conditions are shown to facilitate a comparative analysis with the performance results of CSENet across all generations. As shown in Fig. 5, for the UC Merced dataset, the performance of the initial population has outperformed the "Baseline" for both inductive and transductive inference, which verifies that the self-supervised tasks can improve the accuracy. In addition, the help of the pretext task combination is different on diverse datasets and even inconsistent on different inference approaches. The above results demonstrate

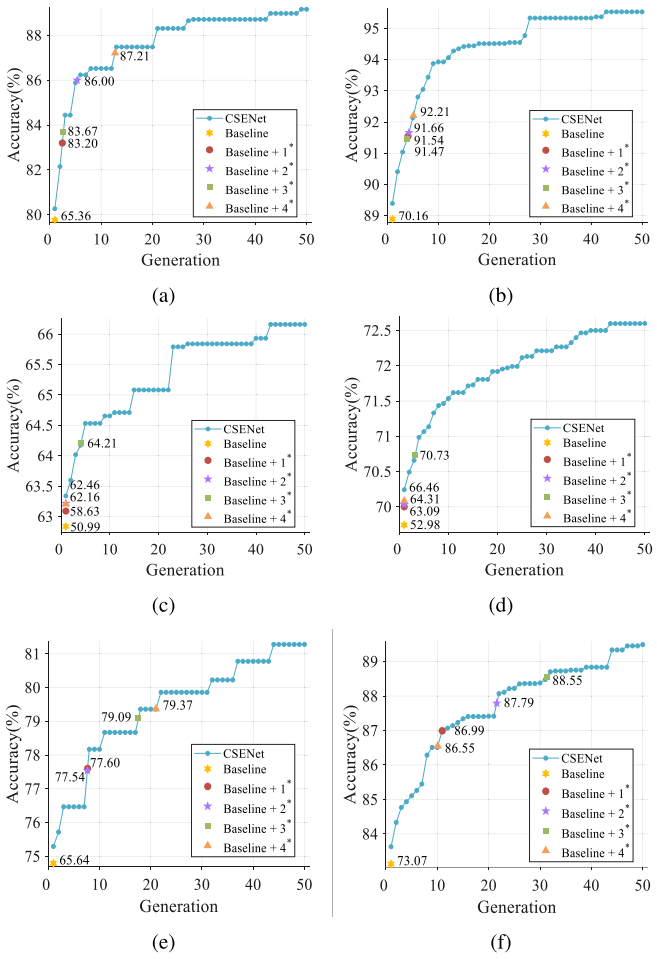


Fig. 5. Accuracy convergence and best results with the different number of pretext tasks for one-shot complete training. 1*, 2*, 3*, and 4* represent the best results obtained after adding 1–4 pretext tasks, respectively. (a) and (b) Inductive inference and transductive inference on the WHU-RS19 dataset, respectively. (c) and (d) Inductive inference and transductive inference on the UC Merced dataset, respectively. (e) and (f) Inductive inference and transductive inference on the NWPU-RESISC45 dataset, respectively.

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TABLE VII
FINAL COMBINATION WEIGHTS OF PRETEXT TASKS ON DIFFERENT DATASETS

Datasets	ω_{GCL}	ω_{SCL}	ω_{RP}	ω_{MM}
WHU-RS19	0.8744	0.7544	0.5014	0.9251
UC Merced	0.8644	0.0392	0.6339	0.5645
NWPU-RESISC45	0.6052	0.3579	0.6814	0.4569

that the optimal combination of pretext tasks varies with tasks and data, and confirm the importance of autonomous search for collaborative effect. In addition, the horizontal coordinate of the comparison method is indexed by the number of generations required for CSENet to achieve the same level of accuracy, which can intuitively show the generation that the proposed method outperforms the comparison method. From Fig. 5, it can be seen that the final accuracy of CSENet is better than the comparison method, which indicates that the proposed CSENet can improve performance through the pretext task collaboration.

The optimal combination weights searched by the CSENet on different datasets are displayed in Table VII. It can be seen that although all the auxiliary tasks are participated in the collaboration, the weights of each task are not the same. For example, in the UC Merced dataset, the corresponding weight for SCL is small, which indicates that for relatively simple scenarios, local spatial information plays a less important role. Besides, the results of the combination with different weights also verify that the pretext tasks do not assist the main task to the same extent. In addition, the table shows that the final combination weights obtained from the three datasets are not the same. For instance, the weight distribution of the four auxiliary tasks in the NWPU-RESISC45 dataset is balanced, while the weight distribution under the other two datasets is significantly biased, which demonstrates that the collaboration of the self-supervised tasks changes with different remote sensing data. The results of the experiments also support the significance of adaptive exploration of task collaboration, and verify that adaptive collaboration can help improve the performance and generalization of the model.

V. CONCLUSION

In this article, we propose a novel CSENet framework for few-shot remote sensing scene classification which automatically obtains optimal combinations of SSATs. Based on the automatic search, the framework alleviates the inflexibility of manually selecting task combinations and the difficulty in capturing the relationship between multiple pretext tasks. Through the collaborative effect of different pretext tasks, the framework can improve the generalization and the performance on the downstream few-shot classification. In addition, we design an automatic collaborative evolution strategy that adjusts the weight combination for the pretext tasks through the evolution of individuals in the population. The proposed CSENet supports both inductive and transductive inference, and experimental results on different datasets show that CSENet achieves competitive performance over existing SOTA methods.

In future work, we will explore evolutionary multitasking optimization and evolutionary multiobjective optimization to solve the conflict of different SSATs in few-shot remote sensing scene classification.

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